

Longitudinal Gaps and Readout Connectivity in a Pilot Generative Model of Zero-Degree Calorimeter Showers

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October 2026

Abstract

We test whether agreement in mean energy and occupancy implies agreement in shower structure for a hierarchical neutron generator in an ePIC Zero-Degree Calorimeter simulation. The target is raw deposited energy in 6,790 stored readout channels, without digitization or an analysis threshold. One checkpoint trained on 26,624 showers is compared with Geant4 at 10,000 matched incident conditions spanning 50–250 GeV kinetic energy. Among nonempty showers, the generated sample has 0.25 more active layers on average, but its last active layer lies 4.42 layers farther downstream and its occupied span contains 3.95 more inactive layers. The mean graph-component count rises from 23.42 to 59.39. Bounding the difference in occupied-layer run counts leaves an excess of at least 30.45 components within runs, or 84.6% of the observed 35.97-component increase. Thus empty-layer separations alone do not account for the graph fragmentation. The high-level classifier AUROC of 0.775 exceeds the recorded screening maximum of 0.65. These aggregate-only results concern one training seed and a repeatedly inspected validation sample. Physical segmentation, threshold sensitivity and variation across fits remain unverified; the comparison does not establish calibrated detector performance.

1 Introduction

At the Electron–Ion Collider, a zero-degree calorimeter (ZDC) measures neutral particles emitted close to the outgoing hadron beam. The cited SiPM-on-tile design uses hit energies and positions to reconstruct neutron energy and angle [1]. A surrogate for detailed particle transport must therefore reproduce spatial organization as well as total energy. Here, support is the set of stored channels with positive deposit; occupancy counts its members.

Generative calorimeter studies have used normalizing flows, diffusion, and conditional flow matching with regular-grid, point-cloud, or graph representations [2–6]. CaloChallenge assesses complementary energy, shape, classifier, and computational-cost observables [7, 8]. Work on the ALICE Zero-Degree Calorimeter (ZDC) has explored variational autoencoders, adversarial models, diffusion, flow matching, and mixtures of generative experts [9–13]. Recent ZDC flow studies address output variability [14] and neighboring-channel co-activation through a Jaccard co-activation error [15]. Those models use different detector representations and targets, so their reported numerical fidelities are not direct baselines for the raw channel energies studied here.

We present a diagnostic study of a hierarchical ZDC shower generator that resolves the readout into total deposit, layer activity, and channel support. The graph components below diagnose spatial structure; this study does not measure their effect on reconstruction. We ask whether similar raw-deposit and occupancy means also imply similar longitudinal organization. Here, they coexist with deeper deposits and more inactive layers in the occupied span. We derive an aggregate bound showing that most of the component-count excess remains within occupied-layer runs. This graph diagnostic complements co-activation tests; the general need for correlation-sensitive validation is established in the cited literature.

This aggregate-only diagnostic note uses one checkpoint and 10,000 validation conditions; it tests neither architecture superiority nor detector performance.

2 Simulation target and study population

2.1 Readout target and conditioning

The event condition is the incident-neutron four-vector $p^\mu = (E, p_x, p_y, p_z)$, expressed in GeV. The five network inputs are

$$c = (\log(1 + K_{\text{inc}}/100 \text{ GeV}), u_x, u_y, u_z, \log(E/\text{GeV})), \quad \mathbf{u} = \frac{\mathbf{p}}{\max(\|\mathbf{p}\|, 10^{-12} \text{ GeV})}, \quad (1)$$

where $K_{\text{inc}} = \max(E - m_n, 0)$ and $m_n = 0.93956542052 \text{ GeV}$. For $E \geq m_n$, the two energy features carry the same physical information because $E = K_{\text{inc}} + m_n$; both are inputs to this checkpoint. Direction features vanish at zero momentum, and the clamp is numerical.

The target $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{6790}$ is the raw, nondigitized energy deposited in each valid readout channel, with repeated hits to one ID summed in preprocessing. Geant4 supplies the transport [16]. The event total sums stored deposits; it excludes energy lost outside the readout. No analysis threshold or additional time selection is applied; the production integration window is unknown. We call an event with zero total readout deposit an empty shower.

2.2 Geometry and graph representation

The stored geometry has 400 electromagnetic-calorimeter (ECAL) IDs in layer 0 and 6,390 hadronic-calorimeter (HCAL) IDs in layers 1–64: 100 per layer through 63 and 90 in layer 64. ECAL centroids have 30.3 mm horizontal pitch; HCAL mean layer coordinates have median z spacing 24.9 mm and first-to-last span 1.57 m. The ECAL lies at $z = 35.73 \text{ m}$ in the stored frame (Fig. 1).

Here a channel is a stored ID. For 2,400 HCAL IDs, two to four distinct positions share an ID; their unweighted centroid defines the graph node. This does not establish electronic ganging: intentional grouping and identifier aliasing require the production ID/volume mapping to distinguish. The cited design has individually instrumented staggered tiles [1, 17], but equivalence is unverified. The production geometry tag, materials, transport settings and coordinate origin are undocumented; layer indices are not interaction lengths.

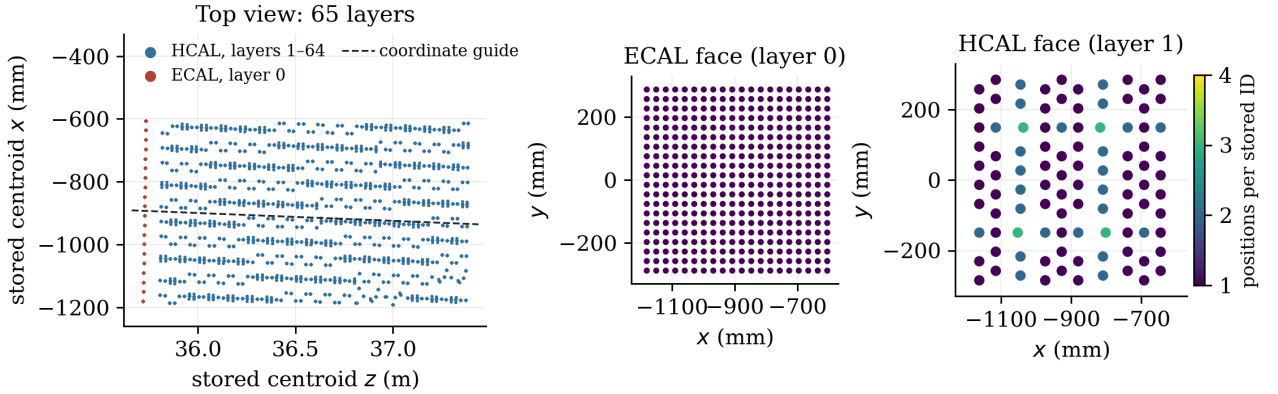


Figure 1: Stored-ID centroids. Left: horizontal projection of 65 layers, with coordinate guide $x = -0.025z$ (unequal axis scales). Center/right: ECAL/HCAL faces, colored by positions per ID. The guide does not establish the beam axis; ID multiplicity does not establish electronic ganging.

The fixed graph gives each channel eight nearest-neighbor links within its layer. For each channel in layers 0–63, it also selects four nearest centroids in the next layer, with each selected longitudinal pair stored in both directions. The resulting directed edge set contains $8 \times 6790 = 54,320$ lateral and $2 \times 4 \times (400 + 63 \times 100) = 53,600$ longitudinal edges, or 107,920 in total. The generator and connectivity diagnostic share this graph, which represents adjacency through readout-centroid distances.

2.3 Training and evaluation populations

The preparation record contains 764,940 showers over 0–300 GeV. A deterministic event-hash split assigns 612,482 to canonical training, 76,158 to validation, and 76,300 to the nominal test partition. The selected checkpoint instead used a pilot subset of 26,624 training and 6,656 validation events; the pilot training population is 4.35% of canonical training.

The primary shower comparison uses 10,000 incident conditions drawn from canonical validation with $50 \leq K_{\text{inc}} \leq 250$ GeV. We refer to this repeatedly inspected validation subset as the development bank D_{dev} . For each four-vector, we compare one stored Geant4 shower with one independently sampled model shower. The pair shares its incident condition; its stochastic shower histories are independent. The development bank’s overlap with the 6,656-event pilot-validation subset cannot be reconstructed from the released aggregates, although the recorded split cross-tabulation places pilot training and validation events within their corresponding canonical partitions. No nominal test event is used here. This reanalysis uses the evaluation summary and static geometry. Per-event masks and energies were not included in that summary; exact run counts, paired structural intervals and threshold studies require a new event-level evaluation of the project checkpoint.

3 Generator and training procedure

The generator builds a shower at three levels of stored readout: event response, longitudinal allocation, and channel deposits (Fig. 2). It separates how much energy is stored (T), where along the detector it appears ($F, \mathbf{A}, \mathbf{B}$), and how it is distributed transversely ($\mathbf{K}, \mathbf{S}, \mathbf{Y}$). Counts precede positions, as in point-cloud models [5, 18]. These observables describe a shower outcome; the cascade does not simulate individual interactions or particle trajectories.

The model has 2,082,507 trainable parameters. A two-block residual multilayer perceptron encodes the five inputs in Eq. (1) into a 128-component condition vector h_c . Discrete heads generate visibility, layer activity, channel counts, and support; two conditional flow-matching models generate the continuous layer and channel energy shares from independent Gaussian draws. There is no learned shower encoder or common random event latent. At inference, the incident four-vector is the only event input. The embedding h_c encodes energy and direction for a fixed geometry; its learned coordinates have no assigned physical observables. Random draws represent conditional shower variability, not added electronics noise.

Below, $\ell = 0, \dots, 64$ indexes layers, i indexes valid channels in layer ℓ , and $Y_{\ell i}$ is a stored deposit. A channel is active if $Y_{\ell i} > 0$ and a layer if its budget $B_\ell = \sum_i Y_{\ell i} > 0$. The event total is $T = \sum_\ell B_\ell$.

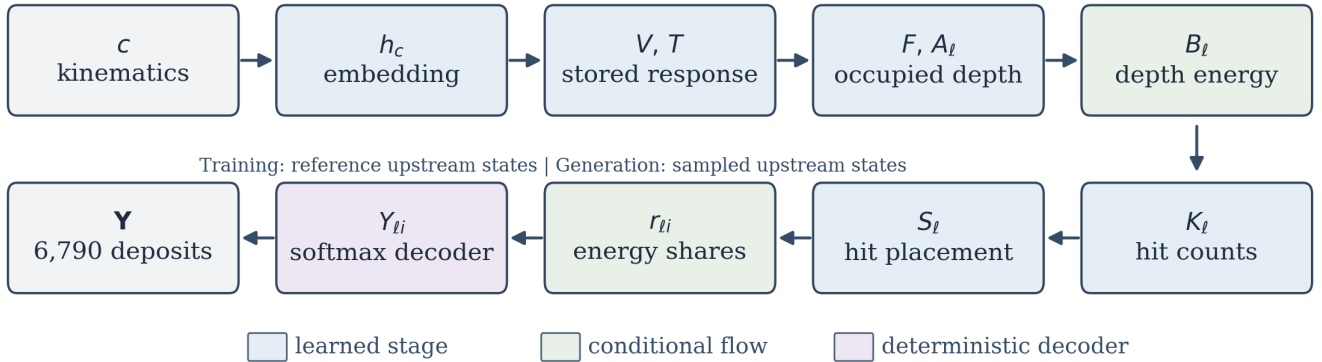


Figure 2: Read the top row left to right, then the bottom row right to left. The cascade maps incident kinematics to stored deposits. Green stages generate energy shares by conditional flows; purple decodes deposits. Arrows show generation order, not every conditioning connection or particle transport. The middle annotation contrasts training and generation inputs.

3.1 Total deposit and longitudinal allocation

Stored response. The response stage separates zero from nonzero stored response (a hurdle model), then describes fluctuations in its magnitude. Here visibility means nonzero deposited energy, not optical visibility or the probability of an inelastic interaction. A Bernoulli head samples a provisional visibility flag V_0 , while a four-component Gaussian mixture describes the transformed total q_T . The Bernoulli logit a_θ , mixture means μ_m , positive widths σ_m , and softmax weights ω_m depend on h_c ; θ denotes the learned parameters and σ the logistic function:

$$\begin{aligned} \Pr(V_0 = 1 \mid h_c) &= \sigma(a_\theta(h_c)), & q_T &= \log(1 + T/10 \text{ GeV}), \\ p_\theta(q \mid h_c) &= \sum_{m=1}^4 \omega_m(h_c) \mathcal{N}(q; \mu_m(h_c), \sigma_m^2(h_c)). \end{aligned} \quad (2)$$

A draw q is mapped back to deposited-energy units, clipped at zero, and capped at the training-derived value $C(K_{\text{inc}})$. The final visibility flag also requires positive incident kinetic energy and a positive candidate deposit:

$$\begin{aligned} C(K_{\text{inc}}) &= \min\{0.630110 \max(K_{\text{inc}}, 1 \text{ GeV}), 61.2338 \text{ GeV}\}, \\ T_{\text{cand}} &= \min\{10 \text{ GeV} \max(e^q - 1, 0), C(K_{\text{inc}})\}, \\ V &= V_0 \mathbf{1}_{K_{\text{inc}} > 0} \mathbf{1}_{T_{\text{cand}} > 0}, & T &= VT_{\text{cand}}. \end{aligned} \quad (3)$$

The cap saturates at 61.2338 GeV above $K_{\text{inc}} \approx 97.2 \text{ GeV}$. Its training-derived tail bound is a numerical safeguard, not a sampling fraction or a mean-response prediction. Clamping a nonpositive inverse-transform value to zero can produce an empty shower even when $V_0 = 1$.

Layer activity. For a visible event, the model draws the first active layer $F \sim \text{Cat}[\text{softmax } f_\theta(h_c, \log(1 + T/\text{GeV}))]$ from 65 categorical probabilities. It forces layer F active and all earlier layers inactive. Separate Bernoulli draws set the later indicators A_ℓ ; conditional on (h_c, T, F) , their joint probability factorizes as

$$p(\mathbf{A} \mid h_c, T, F) = \prod_{\ell > F}^{64} p(A_\ell \mid h_c, T, F, \ell), \quad A_F = 1. \quad (4)$$

F marks the first stored deposit, not necessarily the first inelastic interaction. The mask $\mathbf{A}$ specifies occupied depth and gaps; its factorization omits residual co-activation at fixed (h_c, T, F) . The budget flow then assigns the longitudinal energy profile and ECAL/HCAL partition on that support. A star denotes a reference-derived quantity. For a reference shower, let $B_\ell^* = \sum_i Y_{\ell i}^*$, $T^* = \sum_\ell B_\ell^*$, and $A_\ell^* = \mathbf{1}_{B_\ell^* > 0}$. The target for a nonempty event is a centered log fraction on those layers:

$$a_\ell^* = \log \max\left(\frac{B_\ell^*}{\max(T^*, 10^{-12} \text{ GeV})}, 10^{-8}\right), \quad (x_1)_\ell = A_\ell^* \left(a_\ell^* - \frac{\sum_m A_m^* a_m^*}{\max(1, \sum_m A_m^*)}\right). \quad (5)$$

For an empty reference shower, the target and mask are zero. The centering removes a common log offset that the softmax ignores. Gaussian noise can contain this offset, which final normalization removes. Floors of 10^{-8} and 10^{-12} GeV keep the operations finite; they are numerical conventions, not detector thresholds, and alter small shares. Their activation frequency was not measured.

Energy allocation by flow matching. Both energy-allocation flows learn a velocity field that transports Gaussian noise toward the distribution of reference log fractions, conditioned on the incident particle and upstream shower state. Flow time is an artificial interpolation coordinate, independent of detector time. Straight-path conditional flow matching regresses noise-to-target displacements [19, 20]. For each training example, x_0 is standard-normal noise on the active coordinates, x_1 the reference-derived target, and t is uniform on $[0, 1]$. A mask $M_{bj} \in \{0, 1\}$ identifies active coordinate j of minibatch example b :

$$x_t = (1 - t)x_0 + tx_1, \quad \mathcal{L}_{\text{FM}} = \frac{\sum_{b,j} M_{bj} [v_\theta(x_t, t; \text{ctx})_{bj} - (x_1 - x_0)_{bj}]^2}{\max(1, \sum_{b,j} M_{bj})}. \quad (6)$$

The velocity field v_θ predicts the straight-path displacement $x_1 - x_0$ from the interpolated state x_t , time, and context. Inactive coordinates do not contribute to the masked squared-error loss.

During generation, both flows start from fresh Gaussian noise and use eight masked velocity updates. The context contains outputs sampled by earlier stages, whereas training supplies their reference values. For an active-layer or selected-channel mask M , the implemented rule is

$$z_0 = M \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad z_{n+1} = M \odot \left[z_n + \frac{1}{8} v_\theta \left(z_n, \frac{n+1/2}{8}; \text{ctx} \right) \right], \quad n = 0, \dots, 7. \quad (7)$$

Here $\odot$ denotes elementwise multiplication. Each update evaluates the velocity at the current state z_n and the midpoint *time* $(n + 1/2)/8$, then resets inactive coordinates to zero. It is a first-order explicit update with midpoint-time evaluation, not the midpoint Runge–Kutta method, which also advances the state to the midpoint. For a visible event, the resulting active-layer logits are normalized to fractions π_ℓ , giving budgets

$$\pi_\ell = \frac{A_\ell e^{z_{8,\ell}}}{\sum_{m:A_m=1} e^{z_{8,m}}}, \quad B_\ell = T\pi_\ell, \quad \sum_{\ell=0}^{64} B_\ell = T. \quad (8)$$

Only active layers enter the softmax denominator; their budgets sum to the sampled total and inactive layers receive zero. The layer-flow network uses a 128-component representation and two bidirectional transformer layers with four attention heads. Its context includes h_c , the total T , layer identity, the activity mask, and a sinusoidal flow-time encoding. Bidirectional attention lets each layer use information from all other layers; it is a statistical dependency, not backward particle transport.

3.2 Channel count, support, and deposited energy

Channel counts. Given (h_c, B_ℓ, A_ℓ) and a learned layer identifier, a categorical head draws the number K_ℓ of active channels. A feasibility mask sets $K_\ell = 0$ for inactive layers and restricts active layers to $1 \leq K_\ell \leq N_\ell$, where N_ℓ is the number of valid channels. This models readout multiplicity at a given layer energy. It does not specify particle multiplicity, footprint area, channel positions or a minimum deposit.

Channel placement. The support network uses channel geometry and the sampled layer state to score candidate channels. Its inputs combine standardized (x, y, z) centroids, normalized layer index, ECAL/HCAL indicators, h_c , B_ℓ , and the requested fraction K_ℓ/N_ℓ . Three graph blocks of width 96 propagate information along the fixed directed edge set $\mathcal{E}$. Each block forms messages from neighboring readouts and uses their sum to update the receiving channel. If $h_i^{(r)}$ is the representation of channel i entering block r , and e_{ji} contains five edge inputs (three coordinate differences, distance, and edge type), one block computes

$$m_i^{(r)} = \sum_{j:(j,i) \in \mathcal{E}} \phi_r(h_j^{(r)}, h_i^{(r)}, e_{ji}), \quad h_i^{(r+1)} = h_i^{(r)} + \psi_r(h_i^{(r)}, m_i^{(r)}). \quad (9)$$

Here ϕ_r computes each message and ψ_r computes the residual update. Layerwise averages of the resulting representations pass through a two-layer, four-head bidirectional transformer; its outputs are broadcast back to channels before their final scores $s_{\ell i}$ are formed. Messages let scores depend on nearby readouts; they are statistical comparisons, not particle or energy transport. The support S_ℓ determines transverse placement and therefore graph connectivity. Selection is a separate stochastic step. For valid-channel set $\mathcal{N}_\ell$, the sampler chooses

$$S_\ell = \{i \in \mathcal{N}_\ell : \text{rank}_{\mathcal{N}_\ell}(s_{\ell i}/\tau + g_{\ell i}) \leq K_\ell\}, \quad (10)$$

Rank one denotes the largest noisy score. The sampler uses temperature $\tau = 1$ and independent Gumbel perturbations $g_{\ell i}$ obtained from uniform draws clipped to $[10^{-7}, 1 - 10^{-7}]$ [21]. Gumbel-top- K samples K_ℓ distinct channels without replacement. Selecting the top K_ℓ enforces the requested count exactly, but imposes no adjacency or connectivity constraint.

Channel energies. A second flow divides each layer budget among its selected channels, describing energy concentration and fluctuations on fixed support. It can change energy-weighted shape and centroids, but cannot relocate deposits or repair disconnected support in exact arithmetic. In training, $S_\ell^\star$ denotes the positive-deposit reference channels and $K_\ell^\star = |S_\ell^\star|$. For $K_\ell^\star > 0$, its target is the centered log fraction of each channel’s layer budget:

$$a_{\ell i}^\star = \log \max \left(\frac{Y_{\ell i}^\star}{\max(B_\ell^\star, 10^{-12} \text{ GeV})}, 10^{-8} \right), \quad (x_1)_{\ell i} = \mathbf{1}_{i \in S_\ell^\star} \left(a_{\ell i}^\star - \frac{1}{K_\ell^\star} \sum_{j \in S_\ell^\star} a_{\ell j}^\star \right). \quad (11)$$

Here x_1 refers to the channel-flow target rather than the layer-flow target. The centering preserves the floored relative shares. An inactive layer has zero target coordinates. This flow uses Eqs. (6) and (7) with the reference channel set during training and the sampled set during generation.

The channel-flow velocity network follows the support network’s graph and layer-context design, with the flow state and a sinusoidal time encoding supplied at each channel. Unselected channels are masked before and after every graph block. Eight updates yield logits $r_{\ell i}$ for selected channels; a deterministic softmax then assigns the energy

$$Y_{\ell i} = \begin{cases} B_\ell \frac{\exp r_{\ell i}}{\sum_{j \in S_\ell} \exp r_{\ell j}}, & i \in S_\ell, \\ 0, & i \notin S_\ell, \end{cases} \quad (12)$$

Consequently, in exact arithmetic, $|S_\ell| = K_\ell$, $Y_{\ell i} \geq 0$, $\sum_i Y_{\ell i} = B_\ell$, and $\sum_{\ell i} Y_{\ell i} = T$. These are accounting identities for the model’s sampled readout budget, not conservation of the incident neutron’s full energy. Energy in unrecorded material and leakage are not generated; positivity and closure alone do not establish a physical shower.

3.3 Training and checkpoint selection

Each training component receives reference-derived targets and upstream values (teacher forcing). During generation, it receives their sampled values, allowing errors to propagate through the cascade. We therefore evaluate complete generated showers (free-running generation) in Sec. 5. The joint training objective combines nine weighted losses:

$$\begin{aligned} \mathcal{L}_{\text{joint}} = & w_V \mathcal{L}_{\text{BCE}}(V) + w_T \mathcal{L}_{\text{NLL}}(T) + w_F \mathcal{L}_{\text{CE}}(F) + w_A \mathcal{L}_{\text{BCE}}(\mathbf{A}) + w_B \mathcal{L}_{\text{FM}}^B \\ & + w_K \mathcal{L}_{\text{CE}}(\mathbf{K}) + w_S \mathcal{L}_{\text{BCE}}(\mathbf{S}) + w_R \mathcal{L}_{\text{rank}}(\mathbf{S}) + w_Y \mathcal{L}_{\text{FM}}^Y. \end{aligned} \quad (13)$$

Binary cross entropy (BCE) trains visibility, later-layer activity, and channel-support decisions; categorical cross entropy (CE) trains first-layer and count distributions. The total-deposit negative log likelihood (NLL) uses the density in transformed q_T . Expressing that density in deposited-energy units introduces the Jacobian factor $1/(T + 10 \text{ GeV})$. The ranking term averages $\log[1 + \exp(-(s_i - s_j))]$ over sampled occupied–unoccupied channel pairs. Superscripts B and Y denote the masked layer-budget and channel-share flow losses in Eq. (6); the count loss downweights zeros in inactive layers.

The retained calibration proposal uses training-data gradient norms to set w_j . In that proposal, response, layer-profile, and count weights hit the lower clipping bound; the visibility weight hit the upper bound. The bounds apply before normalization of the weights to mean one. Appendix A records the calibration values and their provenance limits. A weight-sensitivity study would test whether these bounds affect shower structure.

Optimization proceeded through response, profile, count, support, share, and joint stages. The condition encoder was trained with the response head, frozen for the intermediate component stages, and unfrozen for joint training. Each stage inherited the preceding checkpoint; freezing the encoder kept the inputs to previously trained heads fixed during intermediate stages.

Training used AdamW with a peak learning rate of 3×10^{-4} , batches of six accumulated over four steps (effective batch size 24), gradient clipping at one, and seed 20260723. Epoch 90 minimizes the teacher-forced validation objective across 104 retained rows spanning epochs 11–114 and defines the checkpoint analyzed below. Selection used this objective; the free-running support diagnostics were evaluated subsequently. The checkpoint need not minimize those free-running discrepancies.

4 Evaluation protocol

We examine the generated readout at three scales: total and section deposits, longitudinal layer structure, and channel-level support. Response measurements include empty showers and are summarized both over the full sample and in eight 25 GeV incident-energy bins. Numerical checks test the generated deposits against the model’s sampled budgets, while distributional comparisons assess agreement with Geant4.

Support measurements use a strictly positive energy definition; no detector or analysis threshold is applied. For a nonempty shower, let F and L be the first and last active layers, and A_ℓ the activity indicator. The occupied span is $L - F + 1$ and the number of inactive layers within it is

$$G = L - F + 1 - \sum_{\ell=0}^{64} A_\ell. \quad (14)$$

This counts inactive layers, not contiguous runs. An interior gap occurs when $G > 0$. We also count active channels and weakly connected components $C_1, \dots, C_m$ of their induced subgraph, ignoring edge direction, and report the largest-component fraction $\max_j |C_j|/|S|$ for occupied set S . A component is a group of occupied channels joined by paths using only occupied channels. These support metrics condition on nonempty readout, leaving 9,907 reference and 9,858 generated showers. The two nonempty subsets need not contain the same incident conditions.

The component count depends on longitudinal activity even if within-layer support were unchanged. Graph edges join only the same or adjacent layers. If R is the number of maximal runs of occupied layers, then $m \geq R$: an empty layer between two occupied runs blocks every path between them. Consecutive empty layers constitute one separation, so G alone specifies neither R nor lateral fragmentation. For example, occupied layers 1, 2, 5 have span five, $G = 2$, and $R = 2$, whereas 1, 3, 5 has the same span and G but $R = 3$. Define $Q = m - R$, the component count beyond one per occupied-layer run (Fig. 4). Writing $I = \mathbf{1}_{G>0}$ gives $1 + I \leq R \leq G + 1$: any interior gap requires at least two occupied runs. Thus $\overline{m} - 1 - \overline{G} \leq \overline{Q} \leq \overline{m} - 1 - f$, where $f = \overline{I}$ is the nonempty gap fraction. Subtracting the separate sample bounds gives

$$\Delta \overline{m} - \overline{G}_{\text{gen}} + f_{\text{ref}} \leq \Delta \overline{Q} \leq \Delta \overline{m} + \overline{G}_{\text{ref}} - f_{\text{gen}}, \quad \Delta = \text{generator} - \text{reference}.$$

This deterministic aggregate bound is not a confidence interval or causal decomposition. With event activity masks, R and Q could be computed exactly; the aggregate report does not retain them. Components characterize graph-defined support, not detector-reconstruction clusters [22].

For all-event active-channel counts, including empties, the one-dimensional Wasserstein distance W_1 is the area between empirical cumulative distributions, in channels [8]. The 95% percentile intervals for W_1 , the zero-deposit fraction difference and the mean normalized response residual use 1,000 bootstrap resamples, stratified by incident-energy bin and preserving matched pairs. Intervals for the energy-bin differences, mean longitudinal profile, gaps, and graph components are unavailable. All quoted intervals describe resampling variation within the development bank.

A classifier two-sample test (C2ST) distinguishes generated from reference showers; AUROC (area under the receiver operating characteristic curve) has chance benchmark 0.5 [8, 23]. The original three-seed battery uses a row-wise split. Members of a matched condition pair can enter opposite partitions; its condition-only AUROC of 0.464 is a failed control consistent with opposite-label twin bias. A separate pair-grouped monitor records the same run and epoch (dicos-f-02, epoch 90), with condition-only/high-level AUROCs 0.500/0.893. Its different event sample precludes a direct comparison. Appendix B gives the seeds and recovered evaluator specification.

5 Results

Table 1 compares deposits over all 10,000 conditions and support over separately nonempty samples. Similar mean deposit and activity coexist with larger longitudinal spans, gap counts and graph-component counts.

Table 1: Aggregate comparison on the development bank. Deposit means and zero-deposit counts use all 10,000 matched conditions; support means use nonempty showers (9,907 Geant4 and 9,858 generated). Differences are generator minus reference, computed before rounding; pp denotes percentage points.

Quantity	Geant4 reference	Generator	Gen. – ref.
Zero-deposit events	93 (0.93%)	142 (1.42%)	+49 (+0.49 pp)
Mean total deposit (GeV)	4.324	4.369	+0.045
Mean ECAL deposit (GeV)	2.100	2.275	+0.175
Mean HCAL deposit (GeV)	2.224	2.094	−0.129
Mean first active layer	0.510	0.735	+0.225
Mean last active layer	56.20	60.62	+4.42
Mean active layers	54.58	54.83	+0.25
Mean occupied span	56.69	60.88	+4.20
Mean inactive layers in span	2.10	6.05	+3.95
Events with an interior gap	52.43%	93.53%	+41.10 pp
Mean active channels	1617.6	1588.8	−28.7
Mean weak graph components	23.42	59.39	+35.97
Mean largest-component fraction	94.90%	88.87%	−6.03 pp

5.1 Deposited energy and empty showers

The all-event mean total deposit is 4.369 GeV for generated showers and 4.324 GeV for Geant4, a difference of 0.045 GeV (1.0%). The similar total means conceal opposing section-level shifts: generated ECAL deposit is higher by 0.175 GeV (8.3%) and HCAL deposit lower by 0.129 GeV (5.8%) (Table 1). Figure 3 shows their mean longitudinal allocation. At the reference HCAL maximum (layer 9), the generated mean is 8.6% lower. Mean profiles cannot reveal individual interior gaps.

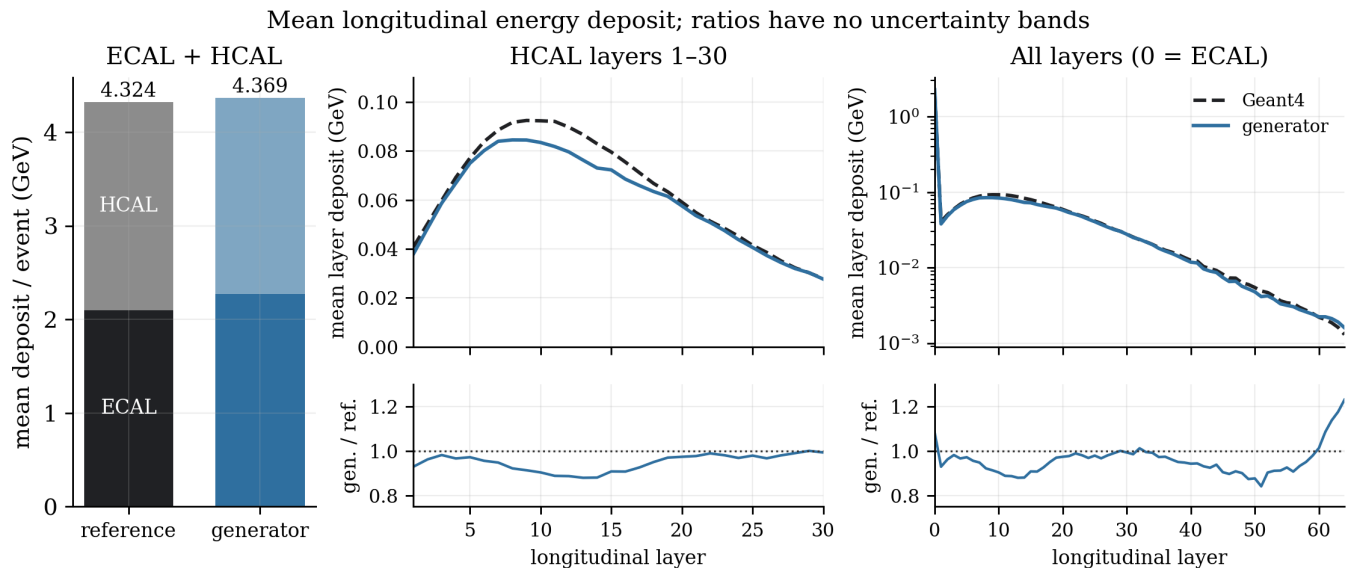


Figure 3: Mean deposited energy over all 10,000 development-bank events. Left: ECAL and HCAL deposits expose section-level compensation. Center: HCAL layers 1–30 (linear scale). Right: all layers, including ECAL at 0 (log scale). Dashed curves denote Geant4, solid curves the generator. Lower panels show ratios of means; uncertainty bands are unavailable. Empty showers are included.

Table 3 gives binwise means and widths; the pooled standard deviations are 4.759 and 4.830 GeV. The mean paired residual $(T_{\text{gen}} - T_{\text{ref}})/K_{\text{inc}}$ is 2.94×10^{-5} , with 95% bootstrap interval $[-8.51, 9.34] \times 10^{-4}$, which includes zero. This tests a different, energy-normalized quantity from the 0.045 GeV mean difference; it does not supply that difference’s paired covariance.

The model produces 142 empty showers versus 93 in Geant4, a difference of 49 events or 0.49 percentage points. The paired-bootstrap 95% percentile interval for this fraction difference is 0.19–0.76 percentage points. The corresponding nonempty samples contain 9,858 generated and 9,907 reference showers. The report’s visibility labels do not retain provisional V_0 , so the 142 generated zeros cannot be split reliably into hurdle and clamp contributions.

5.2 Longitudinal activity

The generated showers have nearly the same mean number of active layers as the reference (54.83 versus 54.58), but the occupied layers lie farther downstream. Their mean first active layer shifts by +0.225 and last active layer by +4.42. The nonempty fraction starting in ECAL is 92.53% versus 93.81%. The occupied span grows by +4.20 layers while the count of active layers grows by only +0.25, leaving +3.95 inactive layers inside the span. The fraction of nonempty showers with an interior gap rises from 52.43% to 93.53% (Table 1). For nonempty showers, mean energy-weighted depth, $\sum_{\ell} \ell B_{\ell} / \max(T, 10^{-9} \text{ GeV})$, is instead 14.14 for the generator and 14.15 for Geant4. Including empty events as zeros gives 13.94 and 14.02, respectively. This distinguishes occupied extent from energy-weighted depth, but does not determine the energy carried by disconnected components. Mean energy in layers 57–64 is 17.85 MeV for the generator and 17.19 MeV for Geant4, about 0.4% of each total. This fixed-region average does not isolate the energy of the additional occupied layers or disconnected components.

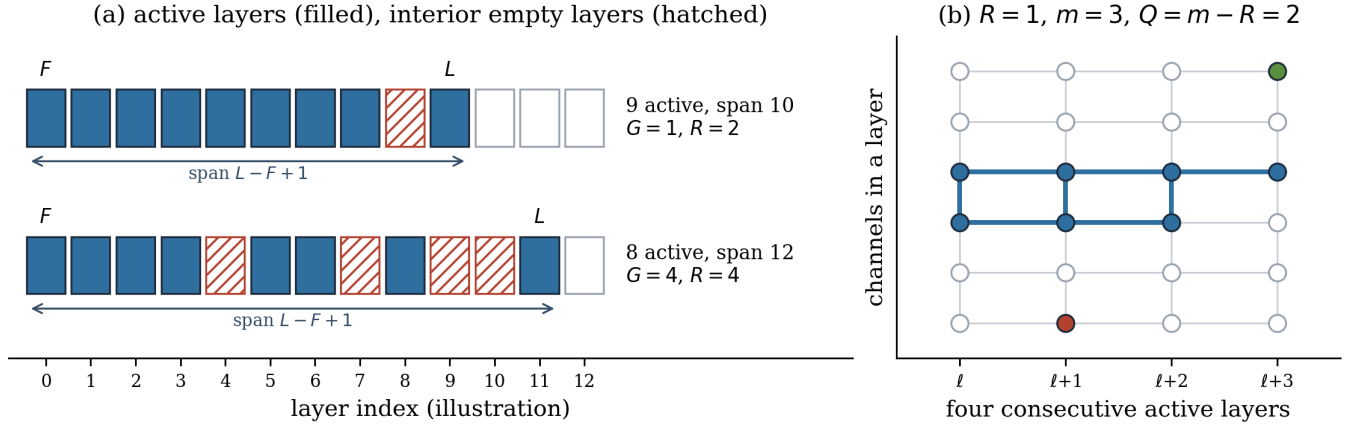


Figure 4: Definitions on an illustrative graph, not shower data. Left: active layers (filled), interior empty layers (hatched), first and last layers F, L , gap count G and occupied runs R . Right: all four layers are active ($R = 1$), but occupied channels form three components ($m = 3$), so $Q = m - R = 2$. Pale edges join adjacent nodes of this toy grid; the detector graph is defined in Sec. 2.

5.3 Channel support and graph structure

Generated showers have 28.7 fewer active channels on average, but 35.97 more weak components on the model graph. The largest component contains 88.87% of occupied channels on average, versus 94.90% in the reference. The bound in Sec. 4 places the mean within-run excess between 30.45 and 37.14 components; at least 84.6% of the component increase cannot be explained by additional runs. The all-event active-channel-count distribution has $W_1 = 58.68$ channels, with a paired-bootstrap 95% interval of 51.47–66.92 channels. This distance includes the zero-count mass, whereas the means above condition on nonempty showers. Two

retained reference-half splits give 13.78 and 15.82 channels (5,000 events per half). These are finite-sample comparison scales, not critical values or matched-size significance estimates.

The fraction of directed graph edges whose endpoints are both occupied averages 0.1375 for the generator and 0.1458 for Geant4 over all 10,000 events. This measures local co-occupancy on the same graph, but is not normalized for occupancy and does not isolate a correlation discrepancy.

5.4 Classifier and numerical checks

The original C2ST battery gives mean AUROCs of 0.775 for high-level shower summaries, 0.791 for channel energies and 0.856 for layer profiles across three classifier seeds. Its high-level score exceeds the recorded screening maximum of 0.65. The failed condition-only control prevents a calibrated fidelity interpretation; a pair-grouped rerun on the same bank is needed. These scores and the failed criterion remain as provenance, not percentages of physically incorrect showers.

The evaluator uses a common event/layer residual tolerance: the larger of 2×10^{-5} GeV and 10^{-5} times the largest sampled event total in each batch. All 1,250 batches of eight passed, with no nonfinite/negative deposit or invalid support/count/mask. The zero support-mask mismatch verifies that FP32 decoding created no selected-but-empty channels in this sample; such losses would bias support metrics. Maximum event and layer residuals were 1.14×10^{-5} and 3.05×10^{-5} GeV. The maximum layer residual exceeds the earlier fixed 2×10^{-5} GeV criterion; the reported pass applies specifically to the batch-dependent criterion.

6 Discussion

6.1 Interpretation and diagnostic tests

Graph connectivity distinguishes more than empty-layer separations. The bound in Sec. 4 gives $\overline{Q}_{\text{gen}} \geq 52.34$ and $\overline{Q}_{\text{ref}} \leq 21.89$, hence $\Delta\overline{Q} \geq 30.45$: at least 84.6% of the observed 35.97-component excess remains beyond one component per occupied run. This establishes a within-run graph-fragmentation discrepancy in these samples. It does not identify a purely lateral effect: adjacent occupied layers may also fail to connect through the selected edges. Nor does it assign a causal percentage to a model stage. The separately nonempty populations need not share the same incident conditions.

Equation (4) assumes conditional independence of later-layer activity given (h_c, T, F) . In a three-layer example, fix $F = 0$ and let layers 1 and 2 be jointly active with probability p . Independent draws with the same marginals preserve the mean active-layer count but give an interior-gap probability $p(1 - p)$ instead of zero and increase the mean last layer by $p(1 - p)$. This explains a possible signature, not the measured magnitude. Sampled T and F can still induce dependence when their values vary; the first deposited layer is not necessarily the first inelastic-interaction depth.

A reference activity null controlling energy, direction, T and F would test whether factorization reproduces the depth and gap shifts. An oracle cascade must replace the full activity mask; reference T and F alone leave the factorization intact. L2LFlows and iCaloFlow model interlayer dependence [24, 25] and motivate a matched comparison.

BCE-trained logits are not automatically calibrated inclusion probabilities after fixed-count Gumbel selection. Independent channel perturbations can disrupt connected support even with smooth scores. Occupancy calibration conditional on layer and count, and a matched-marginal placement null, would test that hypothesis. The channel flow cannot repair selected support in exact arithmetic. Shared random upstream states and cross-layer attention can nevertheless induce centroid correlations without a common event latent; measuring those correlations would test shower-axis coherence.

6.2 Limitations and robustness

Paired sampling intervals for the headline structural means are not retained. Large descriptive differences do not establish robustness across training seeds, thresholds or graphs. A three-seed study and full-data fit

are needed to separate architecture effects from training-population and optimization effects.

Strict-positive support is a property of stored simulation deposits, not electronics readout. Soft or late deposits can change gaps and components; their contribution cannot be quantified without channel-energy spectra and time information. Detector-specific threshold and integration-window studies, and an independently defined physical-neighbor graph, are needed before interpreting the graph discrepancy as detector performance. The cited design uses a 0.5 MeV MIP scale, a reconstructed-hit cut above 0.5 MIP and $t < 275$ ns [1]. These motivate a 0.25 MeV HCAL sensitivity scan, but transferring the cuts requires the production segmentation and timing convention; a reference-only time cut would compare different targets.

The incident four-vector omits an explicit impact position. Whether direction determines that position depends on the production vertex and recording surface, which need verification. Position-conditioned residuals and the joint empty-shower contingency table would test whether conditioning and geometric acceptance are learned; marginal empty fractions alone cannot answer this.

Training loss averages updating minibatches, whereas validation uses the epoch-end model in the available implementation; historical runtime identity is unverified. Response-tail studies require counts of generated draws clipped by the cap and reference deposits above it. Numerical sensitivity requires a solver-step study. A locked bank with verified identities and repeated generation at fixed conditions is also needed.

6.3 Computational performance and detector applications

The evaluation took 3,442 s for 10,000 pairs. Its nonoverlapping timed analysis stages total 1,009 s, leaving 2,433 s (0.243 s per event) for generation, I/O and other untimed work; generation latency was not isolated. A matched end-to-end benchmark must include both solvers, decoding and output handling, with hardware and batch size specified for generator and Geant4.

Relating this raw-deposit study to detector performance requires production metadata, training-only calibration applied unchanged to both samples, and energy and position reconstruction. The ECAL and HCAL shifts cancel only in the unweighted stored-energy sum. A sampling fraction from another production cannot establish this model’s calibrated bias.

7 Conclusion

This pilot comparison shows that similar mean energy and occupancy do not imply matched longitudinal or graph structure. A difference of 0.25 active layers coexists with a last active layer shifted by 4.42 layers and 3.95 additional inactive layers in the occupied span. At least 30.45 of the 35.97 excess components remain within occupied-layer runs. Empty-layer separations are therefore insufficient to explain fragmentation on this graph. Connecting this descriptive result to detector performance requires conditional activity and support-sampling tests, threshold and graph sensitivity, and reconstruction studies.

Code and data availability

The [project repository](#) provides aggregate reports, geometry, training history, figure scripts and manuscript source; its README identifies revisions and build audits. The collaboration-owned events and checkpoint are not redistributed. Generation, retraining and event-level studies require those resources.

Acknowledgments

The author thanks Dr. Wen-Chen Chang (Institute of Physics, Academia Sinica) for his generous mentorship, guidance, and encouragement, and the Institute of Physics laboratory community for their welcome and support.

Author contributions and use of generative AI

Julian Juan: conceptualization, methodology, software, investigation, visualization, and writing.

OpenAI Codex assisted with software and manuscript drafting, specification review, literature searches, and editing. Anthropic Claude provided an additional manuscript review and revision proposal; selected text and figure designs were incorporated after verification. The author supplied the research ideas, checked the work, and takes responsibility for the scientific content.

A Recorded settings and reproducibility limits

The analyzed checkpoint is `dicos-f-02`, epoch 90, from the `calibrated_lr3e4` family, with training seed 20260723. Its SHA-256 is

491284c7423f365230d34b0443f95aa4888ec770bdc673c4c979897bad8acbce

and its recorded frozen-configuration SHA-256 is

116bc8c220b07ce54ae07196bdd6ed8e835775c8c937182a209a799dc94ae9c5

The selected validation objective is 4.4837676194. The recorded lineage includes a declared learning-rate restart between epochs 70 and 71, followed by annealing; it is not one uninterrupted decay. The retained history covers epochs 11–114. The original runtime configuration, complete component-stage durations, optimizer hyperparameters beyond those in Sec. 3, and exact training software environment are not included in the paper package. These details cannot be reconstructed reliably from aggregate losses alone.

Table 2: Recorded gradient-calibration proposal, rounded to six decimals and ordered as in Eq. (13). These values document the retained calibration record; the historical runtime configuration is unavailable for an independent check of the applied weights.

Weight	Component	Value
w_V	Visibility	2.574417
w_T	Total deposit	0.160901
w_F	First active layer	2.159451
w_A	Layer activity	0.536770
w_B	Layer-budget flow	0.160901
w_K	Channel count	0.160901
w_S	Support BCE	1.324108
w_R	Support ranking	1.477591
w_Y	Channel-share flow	0.444960

The retained calibration values are reproduced by taking inverse median gradient norms relative to their geometric mean, clipping to $[0.25, 4]$, and normalizing the resulting nine weights to mean one. This explains why the final entries in Table 2 can fall outside the clipping interval. Gradient norms were measured on training batches with respect to the shared condition encoder. This is a fixed calibration heuristic, not evidence that the weights are optimal.

The diagnostic report records generation seed 20260723 (the same integer as the training seed, used for a separate sampling run), FP32 precision on a CUDA device, batches of eight, and eight updates for each flow. Its validation-manifest identity is

1bc3a6b21f736fc71551580a93207128c043833c6848627fb16f07475931690a

The appendix tables use the same retained aggregate report as the main text; no new shower generation or test evaluation was performed to produce them.

The canonical split is based on event hashes rather than independently identified simulation jobs. The present study uses validation events only. The broader project previously inspected 40,000 nominal-test

events in an isolated classifier study and a further draw containing 200 test events, with unresolved overlap. Those historical studies did not feed model decisions, but the nominal test partition cannot be described as wholly untouched; 36,100–36,300 events remain uninspected according to the retained record. A count alone does not identify a locked remainder: exact event identities must be recovered before using it.

B Energy-bin results and classifier details

Table 3 resolves the aggregate response comparison into the eight incident-energy bins. The relative mean difference is $100(\mu_{\text{gen}}/\mu_{\text{ref}} - 1)$ and the width difference is $100(\sigma_{\text{gen}}/\sigma_{\text{ref}} - 1)$, where μ and σ are the mean and standard deviation of total deposited energy. Both statistics include empty showers.

Table 3: Deposited-energy response by incident kinetic energy. n is the number of matched conditions; the same n applies to each sample. Means and population standard deviations σ are in GeV; differences are percentages. Bins are lower-inclusive and upper-exclusive, except that the final bin includes 250 GeV (implemented upper edge 250.0001 GeV). Intervals for these differences are unavailable.

K_{inc} (GeV)	n	μ_{ref}	μ_{gen}	Mean diff. (%)	Width diff. (%)	σ_{ref}	σ_{gen}
50–75	1,182	2.398	2.216	−7.59	−6.40	2.830	2.649
75–100	1,270	2.892	2.908	+0.57	+2.80	3.154	3.243
100–125	1,310	3.527	3.513	−0.42	−0.72	3.872	3.845
125–150	1,226	3.849	4.146	+7.74	+9.58	3.921	4.297
150–175	1,288	4.867	4.494	−7.65	−10.89	5.037	4.488
175–200	1,268	5.030	5.320	+5.76	+8.79	4.802	5.224
200–225	1,266	5.921	6.026	+1.78	−1.70	6.127	6.023
225–250	1,190	6.095	6.330	+3.85	+5.24	5.827	6.133

Table 4: All recorded classifier seeds for the original development screening battery. The three seed columns vary the classifier and its row partition; they are not independent generator-training seeds. All three high-level AUROCs exceed the recorded screening maximum of 0.65.

Feature family	20260804	20260805	20260806	Mean
Condition only	0.4739	0.4624	0.4545	0.4636
High-level	0.7789	0.7719	0.7735	0.7748
Channel energies	0.7873	0.7943	0.7914	0.7910
Layer profile	0.8490	0.8580	0.8623	0.8564

The archived evaluator implementation uses a histogram gradient-boosting classifier with a maximum of 100 boosting iterations, maximum depth four, and learning rate 0.08. Its class-stratified random split assigns 14,000 of the 20,000 rows to the fitting partition and 6,000 to the scoring partition; the library may further reserve fitting rows for early stopping under its defaults. The same seed controls this split and the classifier. The split does not group matched condition pairs. These are classifier partitions within canonical validation, not the canonical training and test partitions.

The high-level inputs are total deposit, active-channel count, energy-weighted depth and transverse centroids, radial RMS, largest-channel energy fraction, ECAL fraction, and the energy fraction in layers 48–64. The channel-energy family uses all 6,790 deposits, the layer-profile family uses the 65 layer sums, and the condition-only control uses incident kinetic energy. Where a high-level feature divides by total energy, its denominator has a 10^{-9} GeV floor. Other classifier options follow the library defaults; its exact historical version is not established by the retained report.

These settings were recovered from archived implementation source and do not establish an exact source match to the historical evaluation environment. Source hashes and the aggregate report identify the evidence available for this appendix. The 0.65 maximum is recorded in the retained report; its timing relative to analysis is not independently established here. With this class-stratified 70/30 split, 42% of pairs straddle

partitions in expectation; a scored row has a 70% chance of its opposite-label twin entering the fitting partition (about 4,200 scoring rows). This can bias condition-sensitive scores downward. Table 4 does not replace a pair-grouped evaluation on the same bank.

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